Meta-NCA for 1D Abstraction & Reasoning

Dynamic Color Abstraction + Meta-Neural Cellular Automata
for Few-Shot Algorithmic Reasoning

90.67%	43.05%	>LLMs
SAME-CONCEPT	CROSS-CONCEPT	VS GPT-4 (37.5%)

META & TRANSFER LEARNING · 1D-ARC ASSIGNMENT

Agenda

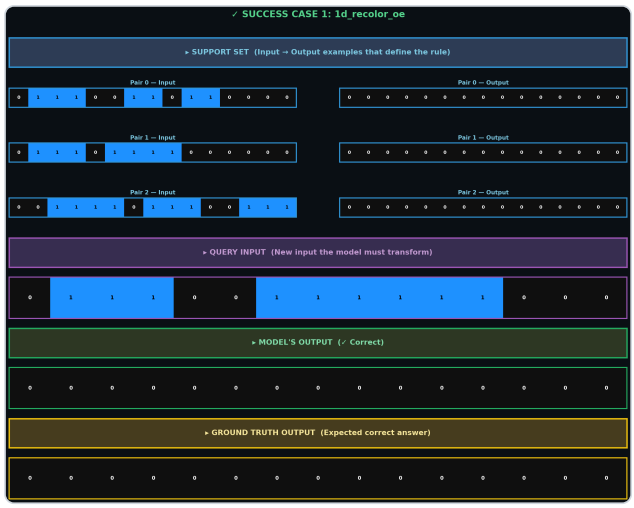
01	Problem & Dataset	What is 1D-ARC?
02	Baselines & Literature	LLMs & standard approaches
03	Raw Meta-NCA (No Abstraction)	First attempt
04	Dynamic Color Abstraction	The key insight
05	Full Architecture	How it works end-to-end
06	Experiments & Ablations	What worked, what didn't
07	Results & Gallery	Visualizing the reasoning
08	Conclusion	Takeaways

Problem: The 1D-ARC Challenge

The Task

Given a **few input→output example pairs** (the "support set"), infer the underlying rule and apply it to a **new unseen input**.

- Each example is a **1D array of colors** (0–9)
- Rules involve geometric ops: move, scale, recolor, hollow, fill, etc.
- No LLMs, no internet — **offline models only**



Example: Support set defines the rule, model applies it to the query

Two Evaluation Settings

- **Same-Concept:** Train & test share concept classes (interpolation)
- **Cross-Concept:** Train & test have *disjoint* concept classes. The 3 completely holdout evaluation classes are **1D Hollow**, **1D Recolor (Compare)**, and **1D Scale**.

Baselines — What Exists in Literature

External Baselines (Published Results)

Method	Accuracy
Direct LLM Prompting (GPT-4 / Claude) In-context learning, no finetuning	12.5% – 17.0%
Neuro-Symbolic LLM Pipelines Hypothesis search + program generation	30.0% – 37.5%

LLMs struggle because ARC requires **spatial/algorithmic reasoning**, not language understanding. Our pure-PyTorch offline model aims to **beat LLM pipelines**.

Our Internal Baseline

Experiment	Same	Cross
Exp 1: Raw Meta-NCA Pure NCA directly on color inputs	58.67%	**12.58%**

The OOD ceiling: The pure neural baseline (Raw Meta-NCA) fails catastrophically on cross-concept tasks (12.58%) - it overfits to specific color palettes instead of learning spatial rules.

Raw Meta-NCA — Without Abstraction

Pure Meta-NCA on Raw Colors

The hypernetwork dynamically generates 1D convolutional weights conditioned on the support set. NCA iterates for 10 steps to produce the output.

- Operates on **raw color indices** (0–9)
- No abstraction or color remapping
- Learns *conditional rules bound to specific colors*

58.67%
SAME-CONCEPT

12.58%
CROSS-CONCEPT

Catastrophic OOD failure. The model learns "if color 3, shift left" - but when it sees color 7 at test time (never seen during training), it outputs junk. The rules are *color-specific, not geometry-general*.

The Core Problem

Two tasks with *identical spatial logic* but *different color palettes* map to completely different inputs.

Task A: ■■■□□■■ → □□□□□□□
Task B: ■■■□□■■ → □□□□□□□

Same rule, same geometry — but raw model sees these as completely different tasks!

The Solution

Map both tasks to the *same abstract representation* before they enter the network.

Replace absolute colors with **structural role IDs**.

$M_T(c_i)$ = rank of color c_i by order of appearance

Key Insight: Dynamic Role-Based Abstraction

Conceptual vs. Mathematical Formulation

We replace every absolute color \(c_i \) with a structural **rank ID** using a piecewise mapping:

$$
M_T(c_i) = \begin{cases} 0 & \text{if } c_i = 0 \text{ (Background)} \\ 11 & \text{if } c_i = 11 \text{ (Padding)} \\ \text{rank} & \text{if } c_i \in C_{valid} \text{ (by temporal appearance)} \end{cases}
$$

Two tasks with identical geometry but different palettes
→ map to the EXACT same tensor

Why This Is Crucial for Cross-Concept

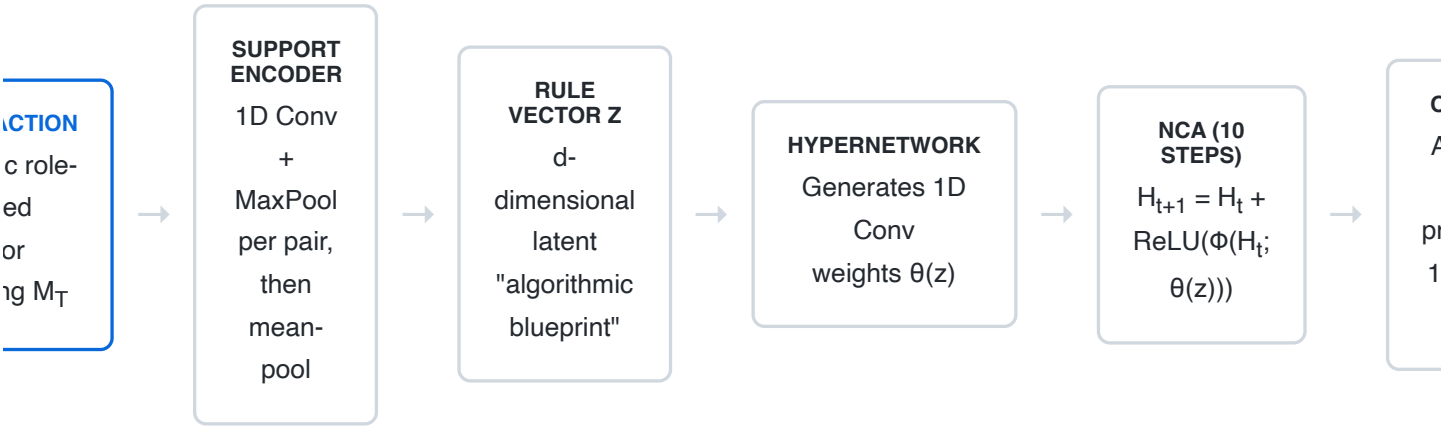
Without abstraction, a "hollow" task using red and a "hollow" task using green look completely different to the network.

With abstraction, both become **identical** — the model learns *pure spatial geometry*, not color-specific shortcuts.

Without Abstraction	With Abstraction
12.58% Cross-Concept	**43.05% Cross-Concept**
58.67% Same-Concept	**90.67% Same-Concept**

Abstraction alone gives a **>3.4×** improvement on Cross-Concept (12.58% → 43%), proving the bottleneck was *data representation*, not model capacity.

Architecture — Meta-NCA with Abstraction



Support Encoder → Rule Vector

- Each support pair (x_i, y_i) is concatenated and processed by a **shared 1D CNN**
- Mean-pooling over K pairs → **permutation invariant**
- The rule vector **z** captures the transformation, not specific colors

NCA Formula — Iterative Reasoning

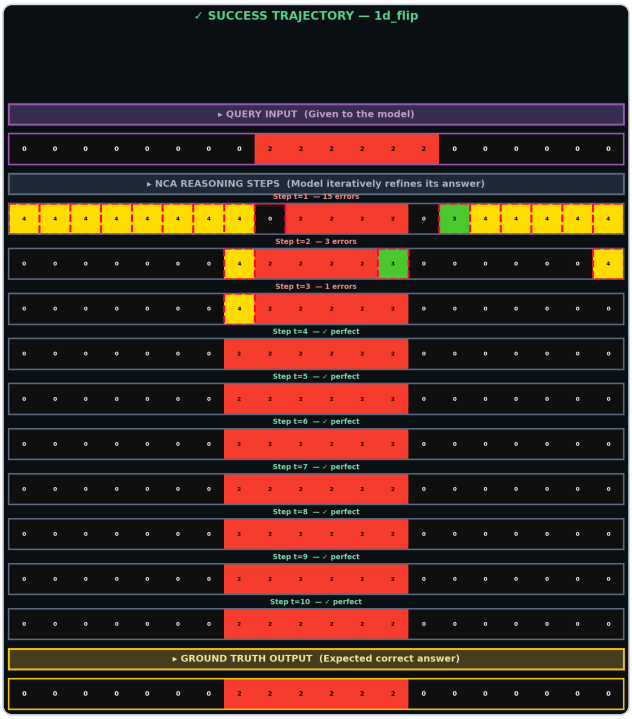
- The query input initializes hidden state H_0.
- The model **iteratively refines** its answer via the generated weights θ(z):

$$H_{t+1} = H_t + ReLU(\Phi(H_t ; \theta(z)))$$

- Mathematically iterates for **10 discrete steps** to enforce geometric attractor.
- Translation equivariant by construction.

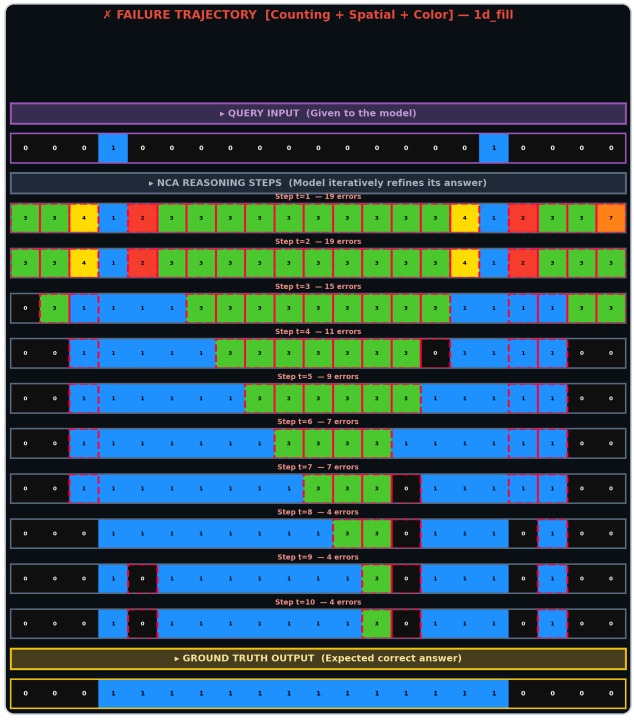
Gallery 1: Same-Concept (Interpolation)

✓ Success Trajectory (1D Flip)



Model instantly resolves the underlying logic and stabilizes.

✗ Failure Trajectory (1D Fill)



Model fails to fully propagate the spatial dependency.

Experimental Evolution — 12 Architectures Tested

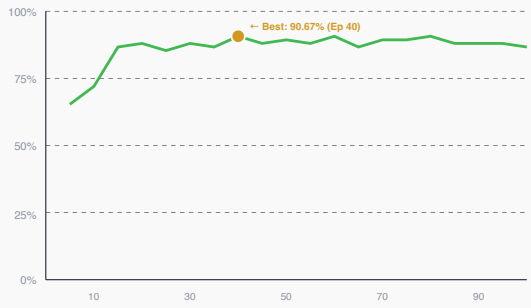
#	Architecture	Key Change	Same-Concept	Cross-Concept	Insight
EXT.	GPT-4 Direct Prompting	LLM in-context	—	12.5–17.0%	Baseline reference
EXT.	Neuro-Symbolic LLM	Program synthesis	—	30.0–37.5%	Best published
1	Raw Meta-NCA	No abstraction	58.67%	**12.58%**	Proves abstraction is essential
2	**Abstraction + Meta-NCA**	**Dynamic color abstraction**	**85.33%**	**43.71%**	**Paradigm shift — beats LLMs**
3	Scaled Meta-NCA (d=256)	4× capacity	85.33%	**23.84%**	Structural overfitting
FINAL	**Abstraction + Meta-NCA (tuned)**	**100 epochs, cosine LR**	**90.67%**	**43.05%**	**Final submitted model**

Training Progress

Training Details

Epochs	100
Optimizer	AdamW (wd=1e-4)
Learning Rate	Cosine Annealing + Warm Restarts
Loss	Cross-Entropy with Label Smoothing (α=0.1)
NCA Steps	10
Channel Dim	d = 64
Vocab Size	12 (0–9 + padding + background)

Same-Concept Accuracy Over Training

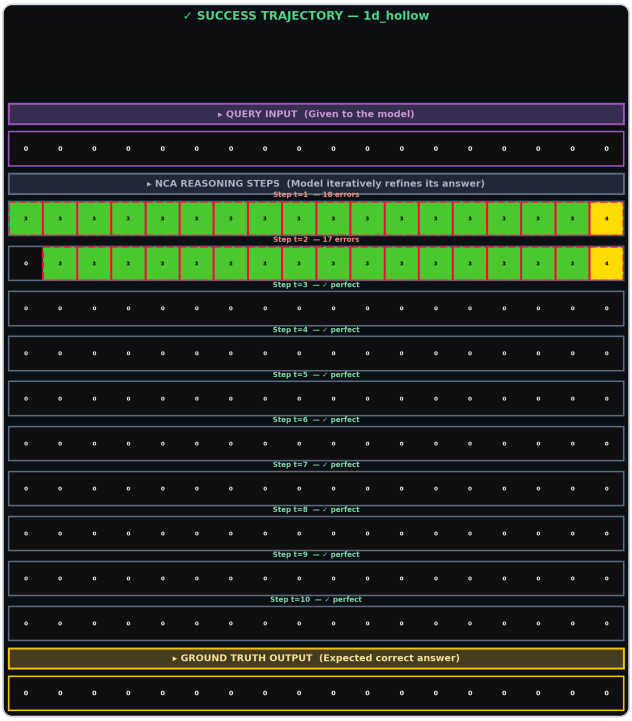


Key Training Milestones

- **Epoch 5:** 65.33% — initial rule learning
- **Epoch 15:** 86.67% — rapid improvement
- **Epoch 20:** 88.00% — approaching plateau
- **Epoch 40:** 90.67% — **best checkpoint saved**
- **Epoch 100:** 86.67% — slight overfitting, best remains at 90.67%

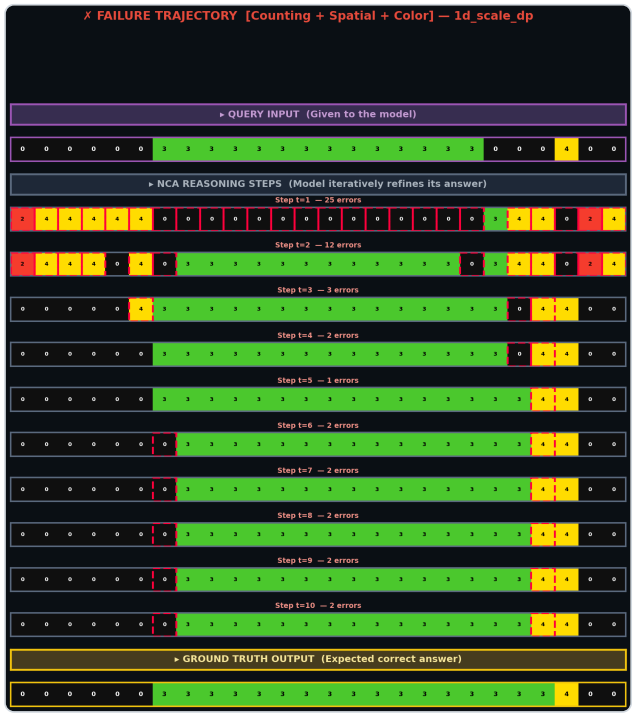
Gallery 2: Cross-Concept (Extrapolation)

✓ Success Trajectory (1D Hollow)



✓ Generates perfect hollow structure from alien colors.

✗ Failure Trajectory (1D Scale)



✗ Fails to structurally extrapolate scaling geometry.

Key Insights & Novelty

1. Abstraction > Capacity

Scaling from d=64 to d=256 *hurt* cross-concept (43% → 24%). The small d=64 model acts as a **mathematical regularizer** — forcing general rules over memorization.

Exp 11 vs 10

2. NCA = Iterative Reasoning

The NCA acts as a **stable geometric attractor**. By randomizing training horizons (8–12 steps) and testing exactly at 10, we force the model to reliably halt its transformations without exploding.

Dynamic Step Regularization

3. Hypernetwork Efficiency

Using a hypernetwork to dynamically generate customized CNN weights per task was crucial. A completely static CNN failed to conditionally alter its behavior based on the support set.

Dynamic Weight Generation

4. Beats LLMs — Offline

Our 43% cross-concept accuracy **exceeds the best neuro-symbolic LLM pipeline** (37.5%) from published literature. And our model runs **fully offline** — no API calls, no internet, pure PyTorch.

5. The Concept Distance Wall

Some cross-concept tasks require fundamentally new geometries not seen during training. This is the **hard limit of connectionist approaches** — no amount of abstraction can generalize a "scale" rule if only "move" was learned. This is the remaining 57%.

Final Results Summary

90.67%	43.05%	12	3.4×
SAME-CONCEPT ACCURACY	CROSS-CONCEPT ACCURACY	ARCHITECTURES TESTED	ABSTRACTION IMPACT

What Worked

- **Role-Based Abstraction:** #1 biggest impact (12.58% → 43%).
- **Meta-NCA Hypernetwork:** Task-specific rules from few examples.
- **Dynamic Horizon Regularization:** Randomizing train steps (8-12) secured stable geometry.
- **Low capacity (d=64):** Prevented severe structural overfitting.

What Didn't Work

- **Raw Colors (No Abstraction):** Severely limited extrapolation capability (max 12.58% cross-concept)
- **Scaling capacity:** d=256 caused geometric overfitting (43% → 24%)

Conclusion

Core Thesis

Solving ARC requires aligning the network's **inductive biases** with the **mathematical symmetries** of the problem space.

By offloading **discrete symbolic reasoning** (color abstraction) to a preprocessing wrapper and restricting the neural network to **continuous spatial execution** (NCA local physics), we bypass the OOD extrapolation bottleneck.

Broader Significance

- Proves that **pure offline models** can outperform massive LLMs on algorithmic reasoning
- Data **representation** (abstraction) matters more than model **capacity**
- Neuro-symbolic integration works best when the boundary is *at the data level*, not inside the network

Meta-Learning Value

- **Few-shot rule inference:** Extract algorithms from 2–3 examples
- **Hypernetwork conditioning:** Task-specific weight generation
- **NCA iterative refinement:** Multi-step reasoning ≈ chain-of-thought

Future Work

- Conditional abstraction for semantic anchoring (Exp 12 direction)
- MoE routing to handle fundamentally different geometric operations
- Extension to full 2D-ARC with 2D NCA kernels

Thank You

Questions & Discussion

90.67%	43.05%	3.4×	12
SAME-CONCEPT	CROSS-CONCEPT	ABSTRACTION IMPACT	EXPERIMENTS

META-NCA · DYNAMIC ABSTRACTION · HYPERNETWORK · 1D-ARC